

QUESTION 1

```
import pandas as pd
import numpy as np

df = pd.read_csv("Dataset_Day2.csv")

unique_names = np.unique(df['name'])
unique_mfrs = np.unique(df['mfr'])
unique_vitamins = np.unique(df['vitamins'])

print("Unique Names:\n", unique_names)
print("\nUnique Manufacturers:\n", unique_mfrs)
print("\nUnique Vitamins:\n", unique_vitamins)
```

```
(myenv) flora_04@Flora:/mnt/d/python/day2$ python3 unique_value.py
Unique Names:
['100% Bran' '100% Natural Bran' 'All-Bran' 'All-Bran with Extra Fiber'
 'Almond Delight' 'Apple Cinnamon Cheerios' 'Apple Jacks' 'Basic 4'
 'Bran Chex' 'Bran Flakes' "Cap'n'Crunch" 'Cheerios'
 'Cinnamon Toast Crunch' 'Clusters' 'Cocoa Puffs' 'Corn Chex'
 'Corn Flakes' 'Corn Pops' 'Count Chocula' "Cracklin' Oat Bran"
 'Cream of Wheat (Quick)' 'Crispix' 'Crispy Wheat & Raisins' 'Double Chex'
 'Froot Loops' 'Frosted Flakes' 'Frosted Mini-Wheats'
 'Fruit & Fibre Dates; Walnuts; and Oats' 'Fruitful Bran' 'Fruity Pebbles'
 'Golden Crisp' 'Golden Grahams' 'Grape Nuts Flakes' 'Grape-Nuts'
 'Great Grains Pecan' 'Honey Graham Ohs' 'Honey Nut Cheerios' 'Honey-comb'
 'Just Right Crunchy Nuggets' 'Just Right Fruit & Nut' 'Kix' 'Life'
 'Lucky Charms' 'Maypo' 'Muesli Raisins; Dates; & Almonds'
 'Muesli Raisins; Peaches; & Pecans' 'Mueslix Crispy Blend'
 'Multi-Grain Cheerios' 'Nut&Honey Crunch' 'Nutri-Grain Almond-Raisin'
 'Nutri-grain Wheat' 'Oatmeal Raisin Crisp' 'Post Nat. Raisin Bran'
 'Product 19' 'Puffed Rice' 'Puffed Wheat' 'Quaker Oat Squares'
 'Quaker Oatmeal' 'Raisin Bran' 'Raisin Nut Bran' 'Raisin Squares'
 'Rice Chex' 'Rice Krispies' 'Shredded Wheat' "Shredded Wheat 'n'Bran"
 'Shredded Wheat spoon size' 'Smacks' 'Special K'
 'Strawberry Fruit Wheats' 'Total Corn Flakes' 'Total Raisin Bran'
 'Total Whole Grain' 'Triples' 'Trix' 'Wheat Chex' 'Wheaties'
 'Wheaties Honey Gold']

Unique Manufacturers:
['A' 'G' 'K' 'N' 'P' 'Q' 'R']

Unique Vitamins:
[ 0 25 100]
```

QUESTION 2

```
import pandas as pd

import pandas as pd
import numpy as np

df = pd.read_csv("Dataset_Day2.csv")

df_HighSodLowProt = df[(df['sodium'] > 100) & (df['protein'] < 3)]

# Print the new dataframe
print("\nDataframe with High Sodium and Low Protein:\n")
print(df_HighSodLowProt)
```

Dataframe with High Sodium and Low Protein:

Dataframe with High Sodium and Low Protein:

		name	mfr	type	calories	protein	fat	sodium	...	sugars	potass	vitamins	shelf	weight	cups	rating
4		Almond Delight	R	C	110	2	2	200	...	8	-1	25	3	1.0	0.75	34.384843
5		Apple Cinnamon Cheerios	G	C	110	2	2	180	...	10	70	25	1	1.0	0.75	29.509541
6		Apple Jacks	K	C	110	2	0	125	...	14	30	25	2	1.0	1.00	33.174094
8		Bran Chex	R	C	90	2	1	200	...	6	125	25	1	1.0	0.67	49.120253
10		Cap'n'Crunch	Q	C	120	1	2	220	...	12	35	25	2	1.0	0.75	18.042851
12		Cinnamon Toast Crunch	G	C	120	1	3	210	...	9	45	25	2	1.0	0.75	19.823573
14		Cocoa Puffs	G	C	110	1	1	180	...	13	55	25	2	1.0	1.00	22.736446
15		Corn Chex	R	C	110	2	0	280	...	3	25	25	1	1.0	1.00	41.445019
16		Corn Flakes	K	C	100	2	0	290	...	2	35	25	1	1.0	1.00	45.863324
18		Count Chocula	G	C	110	1	1	180	...	13	65	25	2	1.0	1.00	22.396513
21		Crispix	K	C	110	2	0	220	...	3	30	25	3	1.0	1.00	46.895644
22		Crispy Wheat & Raisins	G	C	100	2	1	140	...	10	120	25	3	1.0	0.75	36.176196
23		Double Chex	R	C	100	2	0	190	...	5	80	25	3	1.0	0.75	44.330856
24		Froot Loops	K	C	110	2	1	125	...	13	30	25	2	1.0	1.00	32.207582
25		Frosted Flakes	K	C	110	1	0	200	...	11	25	25	1	1.0	0.75	31.435973
29		Fruity Pebbles	P	C	110	1	1	135	...	12	25	25	2	1.0	0.75	28.025765
31		Golden Grahams	G	C	110	1	1	280	...	9	45	25	2	1.0	0.75	23.804043
35		Honey Graham Ohs	Q	C	120	1	2	220	...	11	45	25	2	1.0	1.00	21.871292
37		Honey-comb	P	C	110	1	0	180	...	11	35	25	1	1.0	1.33	28.742414
38		Just Right Crunchy Nuggets	K	C	110	2	1	170	...	6	60	100	3	1.0	1.00	36.523683
40		Kix	G	C	110	2	1	260	...	3	40	25	2	1.0	1.50	39.241114
42		Lucky Charms	G	C	110	2	1	180	...	12	55	25	2	1.0	1.00	26.734515
47		Multi-Grain Cheerios	G	C	100	2	1	220	...	6	90	25	1	1.0	1.00	40.105965
48		Nut&Honey Crunch	K	C	120	2	1	190	...	9	40	25	2	1.0	0.67	29.924285
61		Rice Chex	R	C	110	1	0	240	...	2	30	25	1	1.0	1.13	41.998933
62		Rice Krispies	K	C	110	2	0	290	...	3	35	25	1	1.0	1.00	40.560159
69		Total Corn Flakes	G	C	110	2	1	200	...	3	35	100	3	1.0	1.00	38.839746
0	25	3	1.0	0.75	39.106174											
73		Trix	G	C	110	1	1	140	...	12	25	25	2	1.0	1.00	27.753301
76		Wheaties Honey Gold	G	C	110	2	1	200	...	8	60	25	1	1.0	0.75	36.187559

QUESTION 3

```
df_HighSodLowProt = df[(df['sodium'] > 100) & (df['protein'] < 3)]
"""print("\nDataframe with High Sodium and Low Protein:\n")
print(df_HighSodLowProt)
"""# Group by 'mfr' and calculate average calories
avg_calories = df_HighSodLowProt.groupby('mfr')['calories'].mean()
print("\nAverage Calories by Manufacturer:")
print(avg_calories)
# Find the manufacturer with the highest average calories
max_mfr = avg_calories.idxmax()
max_value = avg_calories.max()
print(f"\nManufacturer with highest average calories: {max_mfr} ({max_value})")
```

Average Calories by Manufacturer:

mfr

G 109.230769

K 110.000000

P 110.000000

Q 120.000000

R 104.000000

Name: calories, dtype: float64

Manufacturer with highest average calories: Q (120.0)